Where the Headroom Lies: Block ℓ_p-Norm Statistics for Cyclic Join Bounds

Extended version: full proofs and additional experiments

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This note accompanies the ICDE 2027 submission and gives the proofs that the paper states in condensed form: the ℓ_p Composition Inequality (the validity of every LpH-CEG bound) and the Cyclic Improvement theorem, and the $bs = 1$ experiments on JOB, a TPC-H-like schema, data size and skew. Notation follows the paper.

1 The ℓ_p Composition Inequality

Setup. A hyperpath $P = (S_0{=}\emptyset, F_1, S_1, \ldots, F_m, S_m)$ adds a block of relations F_j at step j, $S_j = S_{j-1} \cup F_j$, and S_m is the set of all relations. Write $A_j = \mathrm{attr}(F_j)$, $V_j = \mathrm{attr}(S_j)$ ($V_0 = \emptyset$, $V_m = V$), the interface $I_j = A_j \cap V_{j-1}$ and $E_j = A_j \setminus I_j$. Then

(F1) $E_j = V_j \setminus V_{j-1}$, so $\{E_j\}_j$ partitions V;

(F2) $I_j \subseteq V_{j-1}$.

Weights $w_{j,p} \geq 0$ ($p \in \mathcal{P} \subseteq [1, \infty]$) satisfy the attribute-wise cover constraint of the paper: for every attribute $a \in V$, bound at step $j(a)$ (i.e. $a \in E_{j(a)}$),

$$w_{j(a)} + \sum_{j' :\, a \in I_{j'}} s_{j'} \;\geq\; 1, \qquad w_j := \sum_p w_{j,p}, \quad s_j := \sum_p w_{j,p}/p \;\; (1/\infty = 0). \tag{C}$$

A *polymatroid* is $h : 2^V \to \mathbb{R}$ with $h(\emptyset) = 0$, monotone and submodular; $h(X \mid Y) := h(X \cup Y) - h(Y)$.

Lemma 1 (Conditioning)**.** *For a polymatroid h and $Y \subseteq Z$: $h(X \mid Y) \geq h(X \mid Z)$.*

Proof. Submodularity on $X \cup Y$ and Z gives $h(X \cup Y) + h(Z) \geq h(X \cup Y \cup Z) + h((X \cup Y) \cap Z)$, and $h((X \cup Y) \cap Z) \geq h(Y)$ by monotonicity since $Y \subseteq (X \cup Y) \cap Z$. Rearrange. □

Lemma 2 (Fractional subadditivity)**.** *Let f be monotone submodular on subsets of a finite set with $f(\emptyset) = 0$, and let $\lambda_i \geq 0$ satisfy $\sum_{i:\, a \in T_i} \lambda_i \geq 1$ for every $a \in S$. Then $\sum_i \lambda_i f(T_i) \geq f(S)$.*

Proof. Order $S = (a_1, \ldots, a_k)$ and let $S_{<t} = \{a_1, \ldots, a_{t-1}\}$. By monotonicity, the chain rule and Lemma 1, $f(T_i) \geq f(T_i \cap S) = \sum_{a_t \in T_i} f(a_t \mid T_i \cap S_{<t}) \geq \sum_{a_t \in T_i} f(a_t \mid S_{<t})$. Weighting by λ_i and exchanging sums, $\sum_i \lambda_i f(T_i) \geq \sum_t f(a_t \mid S_{<t}) \sum_{i:\, a_t \in T_i} \lambda_i \geq \sum_t f(a_t \mid S_{<t}) = f(S)$, using $f(a_t \mid S_{<t}) \geq 0$. □

Lemma 3 (The obligation)**.** *For every polymatroid h and every w satisfying (C),*

$$\sum_j \Big[s_j\, h(I_j) + w_j\, h(E_j \mid I_j) \Big] \;\geq\; h(V).$$

Proof. Let $g_j = h(E_j \mid V_{j-1})$, $\delta_j = \max(0, 1 - w_j) \in [0,1]$ and $f_j(S) = h(S \mid V_{j-1})$ for $S \subseteq E_j$; f_j is monotone, submodular and $f_j(\emptyset) = 0$.

(i) For each j', the chain rule along the partition (F1) and Lemma 1 give $h(I_{j'}) = \sum_j h(I_{j'} \cap E_j \mid I_{j'} \cap V_{j-1}) \geq \sum_j f_j(I_{j'} \cap E_j)$. Hence $\sum_{j'} s_{j'} h(I_{j'}) \geq \sum_j \sum_{j'} s_{j'} f_j(I_{j'} \cap E_j)$.

(ii) Fix j. If $\delta_j > 0$ then by (C) every $a \in E_j$ has $\sum_{j' : a \in I_{j'}} s_{j'} \geq 1 - w_j = \delta_j$, so the sets $I_{j'} \cap E_j$ with weights $s_{j'}/\delta_j$ fractionally cover E_j, and Lemma 2 gives $\sum_{j'} s_{j'} f_j(I_{j'} \cap E_j) \geq \delta_j\, g_j$. If $\delta_j = 0$ the same inequality holds because the left side is non-negative.

(iii) By (F2) and Lemma 1, $h(E_j \mid I_j) \geq g_j \geq 0$, and $w_j \geq 1 - \delta_j$, so $w_j h(E_j \mid I_j) \geq (1 - \delta_j) g_j$.

Adding (i)–(iii): the left side is at least $\sum_j [\delta_j + (1 - \delta_j)] g_j = \sum_j h(E_j \mid V_{j-1}) = h(V)$ by the chain rule and (F1). □

Theorem 1 (ℓ_p Composition Inequality)**.** *Let D be a database, $\mathcal{Q}(D) \neq \emptyset$, and let $\mu_p(F_j \mid I_j)$ be the ℓ_p-norm of the degree sequence of E_j-values per I_j-value in a relation $R_j \supseteq \pi_{A_j} \mathcal{Q}(D)$—for instance the block join $\bowtie_{R \in F_j} R$ computed exactly on D. If w satisfies (C), then*

$$|\mathcal{Q}(D)| \;\leq\; \prod_{j=1}^{m} \prod_{p \in \mathcal{P}} \mu_p(F_j \mid I_j)^{\,w_{j,p}}.$$

Proof. Let h be the entropy of the uniform distribution on $\mathcal{Q}(D)$; it is a polymatroid and $h(V) = \log_2 |\mathcal{Q}(D)|$. The marginal on A_j is supported on $\pi_{A_j} \mathcal{Q}(D) \subseteq R_j$, so the ℓ_p degree inequality of Abo Khamis et al. (PODS 2024) gives, for $p \in [1, \infty]$, $\frac{1}{p} h(I_j) + h(E_j \mid I_j) \leq \log_2 \mu_p(F_j \mid I_j)$. Multiply by $w_{j,p} \geq 0$, sum, and apply Lemma 3. □

Scope. The proof uses only monotonicity and submodularity, so it holds over the whole polymatroid cone and for every hyperpath, block size and $p \geq 1$. It requires (C) exactly: an LP solver may return weights violating (C) by about 10^{-7}; scaling w by $1/(1 - \varepsilon)$, for a coverage shortfall $\varepsilon = 1 - \min_a(\text{coverage of } a)$, restores (C) and raises a bound B by the factor $B^{\varepsilon/(1-\varepsilon)}$, which grows with B. At the solver's feasibility tolerance ($\varepsilon \leq 10^{-7}$) and the largest cover bound we report ($B < 2^{100}$) the factor is below $1 + 10^{-5}$; the released code applies it; the bounds reported in the paper were computed before it did and sit below their rescaled values by at most that factor. It requires exact statistics of the database being bounded (or deterministic upper bounds on them, since the bound is monotone in each μ); a sampled or sketched statistic, such as the Hutchinson estimate of the common-neighbour ℓ_2 in the paper, is outside the guarantee. When some block join is empty the query is empty and the inequality holds trivially (the released LP gives such a factor weight zero, which only loosens the bound).

Proposition 1 (Necessity)**.** *If nonnegative weights $(w_{j,p})$ make the product above an upper bound on $|\mathcal{Q}(D)|$ for every database D whose atoms' relations are independent (no two atoms are copies of one relation), they satisfy (C). Hence on a fixed hyperpath the cover LP is optimal over all weightings of its statistics that are valid on every such database.*

Proof. Fix $a \in V$ and $N \geq 2$, and let D_N give a the domain $[N]$ and every other attribute a single value, each relation holding all tuples over these domains. Then $|\mathcal{Q}(D_N)| = N$, and the block join of F_j has N tuples if $a \in A_j$ and one otherwise, so $\mu_p(F_j \mid I_j)$ is $N^{1/p}$ if $a \in I_j$, N if $a \in E_j$, and

1 if $a \notin A_j$. The bound is N^{c_a} with c_a the left side of (C) at a, and $N^{c_a} \geq N$ forces $c_a \geq 1$. (The entropy of D_N's join is the modular polymatroid $h(S) = [a \in S] \log_2 N$, so the proposition also holds over the polymatroid cone.) □

Self-joins. The construction sets each atom's relation independently, which a self-join forbids: in $R(a,b) \bowtie R(b,c) \bowtie R(c,a)$ the three atoms share R, D_N is not a database of that query, and indeed weight 3 on the first step's $|R|$ and 0 elsewhere gives the valid bound $|R|^3$ although c is uncovered. Sufficiency (the Composition Inequality) is unaffected, since every self-join database is also a database of the independent query; necessity, and with it the reading that reweighting a fixed hyperpath's statistics is exhausted, holds for independent atoms only.

Why the elementary argument does not suffice. If $w_j \geq 1$ for every step, Lemma 3 follows from (iii) alone. The LP's optima on cyclic queries have $w_j < 1$ (e.g. 1/3 on the closing step of a cycle); the deficit δ_j is paid for by later interfaces, which is exactly step (ii). The cyclic improvement and the need for Lemma 2 have the same source.

Independent checks. An LP over the polymatroid cone decides Lemma 3 at fixed weights; enumerating every vertex of the feasible weight region decides it for all weights. Over 6,197 hyperpath structures on eleven shapes with $|V| \leq 6$ and block sizes 1–3 the minimum slack is 0 to solver precision; an adversarial search over random abstract hyperpath schemes with $|V| \leq 5$ found no violation; and over 207,000 database–block-size checks the bound never fell below the exact count (`scripts/shannon_check_wide.py`, `scripts/validity_search.py`).

2 The Cyclic Improvement theorem

Order the cycle C_k as $R_1(a_1, a_2), \ldots, R_k(a_k, a_1)$ and extend in this order with single-relation blocks. Let $N_j = |R_j|$, M_j the maximum degree and $\mu_p(R_j)$ the ℓ_p-norm of the degree sequence of R_j from a_j to a_{j+1}.

Structure. Step 1 has interface $\emptyset$ and extension $\{a_1, a_2\}$: its degree sequence is the single value N_1. A middle step $2 \leq j \leq k-1$ has interface $\{a_j\}$, extension $\{a_{j+1}\}$ and $\mu_\infty = M_j$. The closing step k has interface $\{a_k, a_1\}$ and empty extension; its conditional degrees are all 1, so $\mu_\infty = 1$ and $\mu_1 = N_k$. With $p = \infty$ and unit weights this gives $B_{\max} = N_1 \prod_{j=2}^{k-1} M_j$.

A feasible point for every common $p \geq 1$. Put weight $\frac{p}{p+1}$ with exponent p on every middle step, $\frac{p}{p+1}$ on step 1 and $\frac{1}{p+1}$ on the closing step with exponent 1. The cover constraints read: a_1: $\frac{p}{p+1} + \frac{1}{p+1} = 1$; a_2: $\frac{p}{p+1} + \frac{p/(p+1)}{p} = 1$; a_i for $3 \leq i \leq k-1$: $\frac{p}{p+1} + \frac{1}{p+1} = 1$; a_k: $\frac{p}{p+1} + \frac{1}{p+1} = 1$. All are tight, so the LP optimum satisfies

$$B_{\mathrm{LP}} \leq N_1^{\frac{p}{p+1}} N_k^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \mu_p(R_j)^{\frac{p}{p+1}} \quad \text{and} \quad \frac{B_{\max}}{B_{\mathrm{LP}}} \geq \Big(\frac{N_1}{N_k}\Big)^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \frac{M_j}{\mu_p(R_j)^{p/(p+1)}}.$$

The exponent must be common (or non-increasing). The constraint of the attribute between middle edges j and $j{+}1$ is $\frac{p_j}{p_j+1} + \frac{1}{p_{j+1}+1} \geq 1$, i.e. $p_{j+1} \leq p_j$. Choosing each edge's best exponent independently is therefore not feasible in general (e.g. $p_j = 1$, $p_{j+1} = \infty$ covers that attribute only to $\frac{1}{2}$); the LP may still mix exponents within a step.

Growth. At $p = 2$ a middle factor $M_j/\mu_2(R_j)^{2/3}$ exceeds 1 exactly when $\mu_2(R_j)^2 < M_j^3$: the sum of squared degrees is below the cube of the maximum degree. A few high-degree values satisfy it; many low-degree ones need not (n values of uniform degree d qualify iff $n < d$, and 10^4 values of degree 1 plus one of degree 20 do not). When every middle factor is at least $c > 1$ the ratio grows geometrically in the number $k - 2$ of middle edges. As $p \to \infty$ every factor tends to 1, and LpH-CEG returns $\min(B_{\max}, B_{\mathrm{LP}})$, so the realized ratio is always at least 1. The boundary factor $(N_1/N_k)^{1/(p+1)}$ does not depend on k; it is 1 for a uniform cycle.

Acyclic queries. On a star the cover constraints force unit weight on every step and the LP returns the max-degree bound. On a path the LP can use an interface at $p = 1$ in place of an extension (on a 4-path it returns weights $(1, 0, 1)$), so acyclicity alone does not force unit weights; what redundant coverage on cycles adds is the slack exploited above.

3 A spectral reading of the common-neighbour statistic (post hoc)

For an even k-cycle of binary relations the count is a trace, $|\mathcal{Q}| = \mathrm{Tr}(A_1 \cdots A_k)$ for the incidence matrices A_i ($v_i \times v_{i+1}$), and the common-neighbour ℓ_2 norm is the Frobenius (Schatten-2) norm of the block product $B = A_i A_{i+1}$. Hölder's inequality for Schatten norms gives, over $m = k/2$ consecutive blocks, $|\mathcal{Q}| \le \prod_j \|B_j\|_{S_m}$, and $\|B\|_{S_m}^m \le \sigma_{\max}(B)^{m-2}\|B\|_F^2$ with $\sigma_{\max}$ bounded above by a Collatz–Wielandt certificate on $B^\top B$. Two numbers per block, both from the pair join ($\sigma_{\max}$ lies outside the polymatroid catalog), give a bound tighter than the polymatroid program on 96/240 long-suite queries (by a median 1.25× there, though 1.11× looser over all) and never below an exact count; with each block's top 24 singular values (Lanczos; uncertified, yet above every exact count) it is tighter on 220/240 and takes IMDb at $|V|$=10 from 14.9× to 3.2× the truth. The next step on the statistics axis may thus be spectral order rather than block size.

4 Which statistics are new: a vocabulary table and a separation

For a binary block $R(a, b) \bowtie S(b, c)$ the statistics the paper uses or compares against are the following ($\deg_R(b)$ is the number of a with $(a, b) \in R$, and $N_R(a)$ the set of b with $(a, b) \in R$).

Statistic	Definition	Pair join?	Used by
single-relation norms	$\|\deg_R(\cdot \mid a)\|_p$, $\|\deg_R(\cdot \mid b)\|_p$	no	LpBound, all programs
pair-join size	$\sum_b \deg_R(b)\deg_S(b)$	no	CorrBound (ℓ_1 inner product)
shared-attribute products	$\sum_b \deg_R(b)^p \deg_S(b)^q$	no	CorrBound's generalized inner products (per its authors)
one-side interface norms	$\big\|\big(\sum_{b\in N_R(a)} \deg_S(b)\big)_a\big\|_p$	no	this paper (join-free)
common-neighbour norms	$\big\|\big(\lvert N_R(a) \cap N_S(c)\rvert\big)_{a,c}\big\|_p$	yes	this paper

Proposition 2 (Separation)**.** *The common-neighbour ℓ_2 norm is not determined by the other rows of the table: there are two databases that agree on every single-relation degree sequence, on the pair-join size, on every shared-attribute product $\sum_b \deg_R(b)^p \deg_S(b)^q$ ($p, q \ge 0$) and on every one-side interface norm, yet differ in the squared common-neighbour ℓ_2 norm (32 against 24), and so in the number of 4-cycles they close.*

Proof. Take R with the 4×4 incidence matrix M and S with $M^\top$, for $M = A$, two disjoint copies of $K_{2,2}$, and $M = B$, the bipartite 8-cycle (row i adjacent to columns i and $i{+}1 \bmod 4$).

Both are 2-regular, so every degree in R and S is 2, every shared-attribute product is $4 \cdot 2^{p+q}$, the pair-join size is 16, and every one-side interface degree is 4. The common-neighbour matrix is $MM^\top$: for A it has eight entries equal to 2 (squared norm 32); for B, four diagonal entries 2 and eight entries 1 (squared norm 24); the largest common-neighbour count is 2 in both. For the 4-cycle $R(a,b) \bowtie S(b,c) \bowtie R'(c,d) \bowtie S'(d,a)$ with $R' = R$, $S' = S$ as separate atoms, the count is $\mathrm{tr}((MM^\top)^2) = \|MM^\top\|_F^2$, again 32 against 24. □

The separation is limited to the statistics listed: it does not claim that no multi-relation term of CorrBound, or any other statistic, determines the common-neighbour norm, and it says nothing about the size of the gain on real data (the paper's ablations measure that).

5 Which statistic needs the join: three-norm details, sketch and selection

Of the eight interfaces of a block of binary relations $R(a,b), S(b,c)$, seven (five non-trivial) are functions of single-relation degree tables (the degree of a is $\sum_{b \in N_R(a)} \deg_S(b)$, that of b is $\deg_R(b) \deg_S(b)$); only $\{a,c\}$, the common-neighbour count $|N_R(a) \cap N_S(c)|$, needs the pair join; its ℓ_2 is determined neither by the other interfaces nor by shared-attribute inner products (Section on the separating example), while its ℓ_1 is the join size. We compute the seven without the join (identical to the materialized statistics on 120/120 Hetionet queries; 0.49 s against 4.6 s per block). With three norms and the flow LP they stop a median 2.3× (max 18.7×) above the full bs=2 bound on the Hetionet long suite (1.5–2.9× on the families, 1.9–57× on the sweep), and the common-neighbour ℓ_2, $\|A_R^\top A_S\|_F$, closes the gap on 12/12 family–seed pairs, 10/10 sweep points, 85/120 Hetionet and 116/120 IMDb queries (Hetionet median 1.00×, max 1.67×); its ℓ_∞ does not (1.70×), and added to bs=1 alone it reaches 1.03× (53/120 exact), so the attribution is order-dependent. In the headline setting (eleven norms, polymatroid) the join-free seven carry a median 88% and 86% of the log-gain on Hetionet and IMDb, the ℓ_2 leaves 1.47× and 1.39× (full on 1/120 each), and the other common-neighbour norms (with the join-free ℓ_1) close the rest. A one-pass triangle-inequality bound on the ℓ_2 recovers nothing on Hetionet (2.2×); a 64-probe Hutchinson sketch lands within 0.91–1.78× of the full bound in 0.32 s per block but is an estimate, not a bound. Post hoc, it picks which joins to pay for: joining the three of a $|V|$=6 query's six blocks whose sketched norm lowers the bound most recovers the full $bs \leq 2$ bound on the median of 34 Hetionet queries (top two on 32 IMDb ones), against half that gain for a random three; timed end to end, this plan takes a median 0.67× the time of joining every block on Hetionet (33/40 cheaper) but 1.39× on IMDb, where join-free statistics, sketches and ranking cost more than the three joins they skip.

6 Which block size to pay for: cost-study details

At bs=1, collecting degree sequences on the four cost-study families costs 22–27 ms against ∼1.9 ms to solve; $bs \geq 2$ multiplies that term by a median 45× there (24× more entries) and 61× (8–558×) on the Hetionet suite, and a cataloged bs=2 estimate costs 10 ms (the paper's long-suite table). Use bs=1 when the bound is re-solved many times per statistics build and $bs \geq 2$ when the pair joins can be materialized ahead of the workload.

Table 1: Unified comparison of guaranteed upper bounds (synthetic power-law cycles, 100V/400E, one instance). "Cover" is our bs=1 LP; "Entropic" is LpBound's bound (via the $2^{|V|}$ program). Lower is tighter. "/CEG" is CEG_M's ratio to Cover and "/Ent" Cover's ratio to the entropic optimum.

Query	AGM	CEG_M	Cover	Entropic	/CEG	/Ent
Cycle-3	8.0K	15.6K	6.8K	5.0K	2.3×	1.36×
Cycle-4	160K	655K	116K	87K	5.7×	1.32×
Cycle-5	3.2M	25.5M	1.97M	1.45M	13.0×	1.36×
Cycle-6	64M	1.12B	31.5M	25.2M	35.7×	1.25×

7 Unified comparison on synthetic cycles

8 Additional $bs = 1$ experiments: JOB, TPC-H-like, data size and skew

On the JOB-medium subset (CI: 3K, MK: 2K, MC: 1K rows) the two co-actor 4-cycles improve 1.94× (with keywords) and 2.80× (with companies) at $bs = 1$ with weights w=0.5, and a star query by exactly 1.00×; the IMDb suite of the paper is the systematic version. On a small TPC-H-like synthetic schema (SF=0.01; not `dbgen` data) every cyclic query we built yields 1.00×; LpH-CEG needs *cyclic structure* together with *degree skew* in a middle relation. On synthetic cycle-4 (10 seeds) the improvement grows with data size, from 5.9±0.7× at 100 vertices to 33.9±1.5× at 2K (relative std <12%; a single-run sweep reaches 57× at 5K), and peaks at moderate skew: 2.3±0.5× at α=0.5, 16.1±0.8× at α=2.0, 6.6±0.4× at α=3.0. At low skew all ℓ_p-norms nearly coincide, at extreme skew ℓ_∞ is already near-tight, and in between ℓ_2 captures the heavy tail.

Every number in this section is produced by the scripts in the repository and checked by its claim audit.

9 Multi-way ordering by per-step residuals

The per-step residuals can order a worst-case-optimal join greedily. As implemented, the residual of a variable prefix is the LP re-solved on the query with the prefix attributes projected out, which bounds the residual for every value of the prefix (and could be computed for LpBound in the same way): GenericJoin iterations drop 2.3× on the Hetionet cycle-4 in our simulator, but over 66 synthetic queries the bound-guided order is 1.3% worse at the median (20 wins, 43 losses; Wilcoxon p=0.003, indicative, as queries share shapes and seeds), so it is a targeted heuristic.

10 Complexity

Collecting $\mu_p(F \mid I)$ for a single relation is one scan, $O(N)$; for a two-relation block it materializes the pair join, except for the interfaces computable from single-relation degree tables (all but the common-neighbour interface of a binary block; paragraph "Which statistic needs the join" of the paper). The cover LP has $m|\mathcal{P}|$ variables and $|V|$ constraints; it solves in milliseconds for the queries in the paper. The flow LP over a $bs \leq 2$ catalog is polynomial in the number of constraints and solves in 0.03–4.4 s at $k = 3$–16 over $4k$ orders.